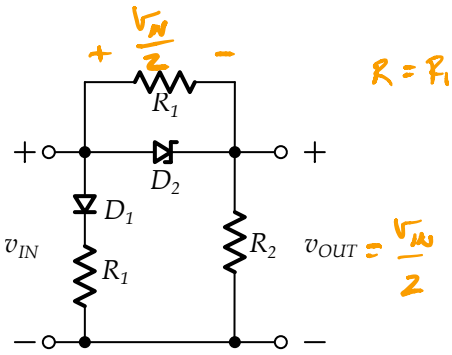
1. (25 points) For the circuit below, assume $V_{D0} = 0.7$ volts, $V_{Z0} = 1.5$ volts, $R_1 = R_2$, and $r_D = r_Z = 0$.



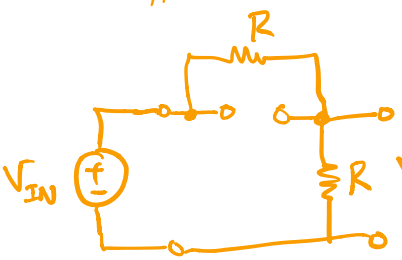
$R = R_1 = R_2$

$v_{OUT} = \frac{v_{IN}}{2}$

(a) Write an expression for v_{OUT} in terms of v_{IN}.

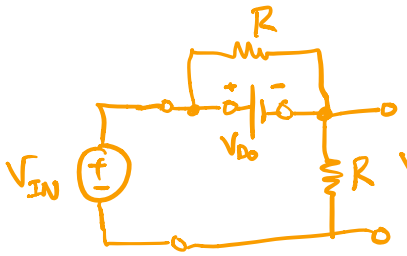
Note that the state of D_1 will not influence the potential at v_{OUT}, so we can ignore that branch for this analysis.

Diode off: $-2V_{Z0} \le v_{IN} \le 2V_{D0}$



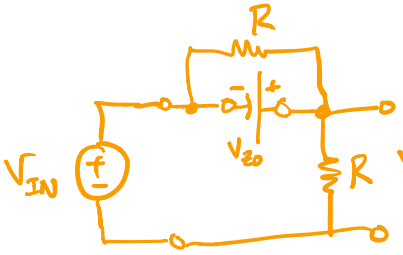
$v_{OUT} = \frac{v_{IN}}{2}$

Forward Biased: $v_{IN} > 2V_{D0}$



$v_{OUT} = v_{IN} - V_{D0}$

Zener Breakdown: $v_{IN} < -2V_{Z0}$



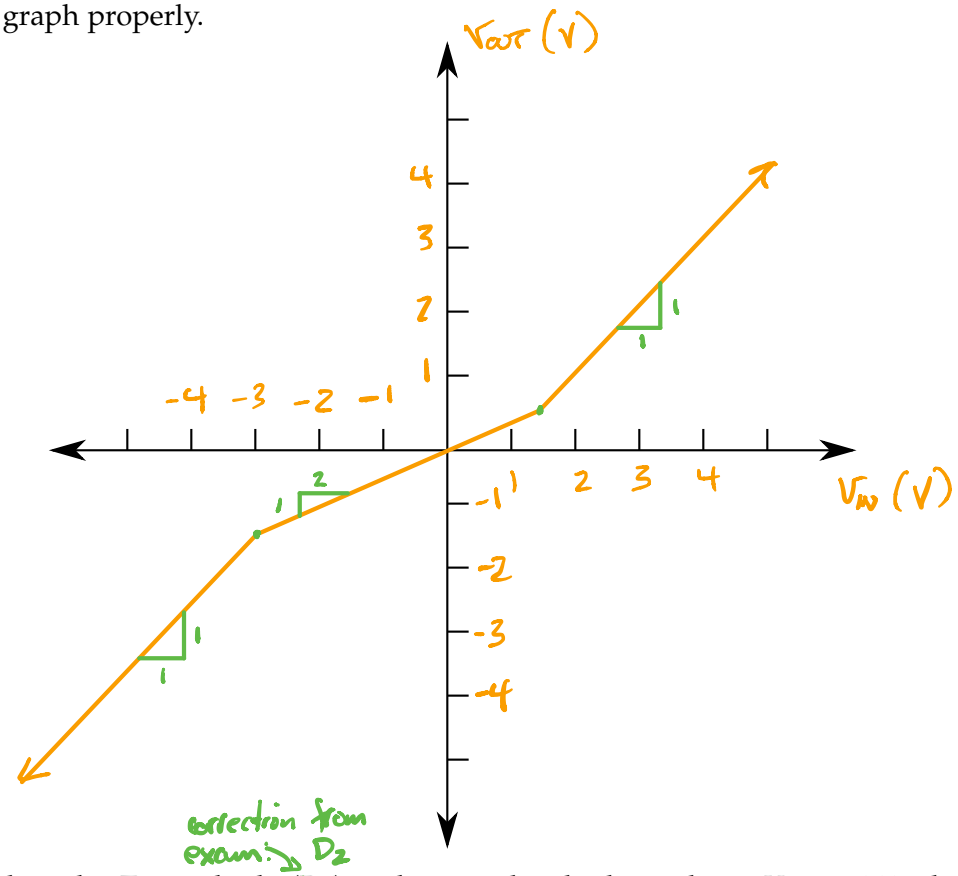
$v_{OUT} = v_{IN} + V_{Z0}$

$$v_{OUT} = \begin{cases} v_{IN} + V_{Z0} & v_{IN} \le -2V_{Z0} \\ \frac{v_{IN}}{2} & -2V_{Z0} < v_{IN} < 2V_{D0} \\ v_{IN} - V_{D0} & v_{IN} \ge 2V_{D0} \end{cases}$$

— OR —

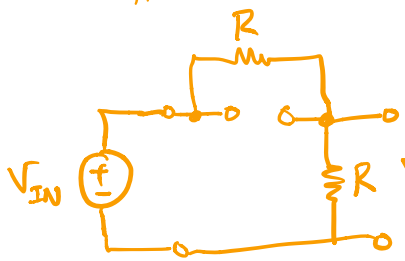
$$v_{OUT} = \begin{cases} v_{IN} + 1.5\,V & v_{IN} < -3\,V \\ \frac{v_{IN}}{2} & -3\,V \le v_{IN} \le 1.4\,V \\ v_{IN} - 0.7\,V & v_{IN} > 1.4\,V \end{cases}$$

(b) Sketch v_{OUT} and v_{IN} for $-4\,\text{volts} < v_{IN} < 4\,\text{volts}$ on the graph provided. Be sure to label the graph properly.



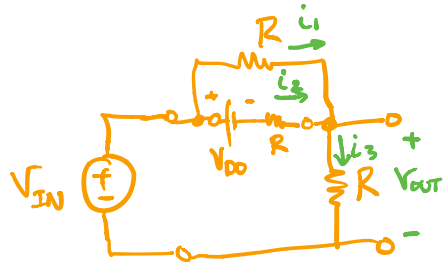
(c) Now, replace the Zener diode (~~D1~~) with a regular diode, and use $V_{D0} = 0.7\,\text{volts}$ and $r_D = R_1 = R_2$. Write an expression for v_{OUT} in terms of v_{IN}.

Diode off: $v_{IN} \le 2V_{DD}$



$$v_{OUT} = \begin{cases} \frac{v_{IN}}{2} & v_{IN} \le 2V_{DD} \\ \frac{2v_{IN} - V_{DD}}{3} & v_{IN} > 2V_{DD} \end{cases}$$

Forward Biased: $v_{IN} > 2V_{DD}$



$$i_1 = \frac{v_{IN} - v_{OUT}}{R} \qquad i_2 = \frac{v_{IN} - V_{DD} - v_{OUT}}{R}$$

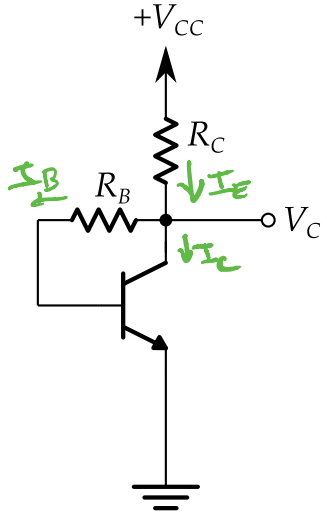
$$i_3 = \frac{v_{OUT}}{R}$$

$$i_1 + i_2 = i_3$$

$$\frac{v_{IN} - v_{OUT}}{R} + \frac{v_{IN} - V_{DD} - v_{OUT}}{R} = \frac{v_{OUT}}{R}$$

$$3v_{OUT} = 2v_{IN} - V_{DD}$$

2. Use the following circuit to complete this problem.



(a) (10 points) Write expressions for V_C and the currents in the transistor in terms of R_B, R_C, V_{CC}, and β.

$V_{CC} = I_E R_C + I_B R_B + 0.7\,V$

$= (\beta+1) I_B R_C + I_B R_B + 0.7\,V$

$$I_B = \frac{V_{CC} - 0.7\,V}{(\beta+1)R_C + R_B}$$

$$I_C = \beta I_B$$

$$I_E = (\beta+1) I_B$$

$$V_C = V_{CC} - I_E R_C$$

- or -

$$V_C = V_{BE} + I_B R_B$$

(b) (10 points) Find values for R_B and R_C so that $V_C = V_{CC}/2$ and $r_\pi = 1\,\text{k}\Omega$ when $V_{CC} = 25\,\text{V}$ and $\beta = 100$.

$$r_\pi = \frac{V_T}{I_B} \Rightarrow I_B = \frac{V_T}{r_\pi} = \frac{0.025\,V}{1000\,\Omega} = 25\,\mu A$$

$$V_C = V_{BE} + I_B R_B \Rightarrow R_B = \frac{V_C - V_{BE}}{I_B} = \frac{12.5\,V - 0.7\,V}{25\,\mu A} = 472\ k\Omega$$

$$V_C = V_{CC} - I_E R_C \Rightarrow R_C = \frac{V_{CC} - V_C}{I_E} = \frac{25\,V - 12.5\,V}{101 \cdot 25\,\mu A} = 4.95\ k\Omega$$

(c) (10 points) Using the values for the resistors and power supplies in the previous part, what is the range of V_C when $50 \leq \beta \leq 150$?

since $I_B = \dfrac{V_{CC} - 0.7V}{(\beta+1)R_C + R_B}$

for $\beta = 50$, $I_B = \dfrac{25V - 0.7V}{(51)(.495\,k\Omega) + 472\,k\Omega} = 0.034\,mA$

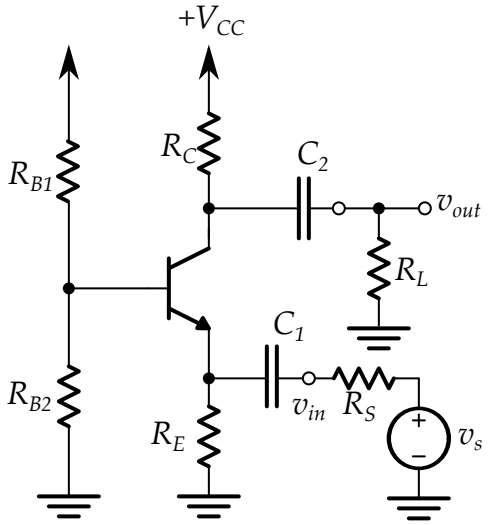
$V_C = 0.7V + (0.034mA)(472\,k\Omega) = 16.5V$

for $\beta = 150$, $I_B = \dfrac{25V - 0.7V}{(151)(.495\,k\Omega) + 472\,k\Omega} = 0.0207mA$

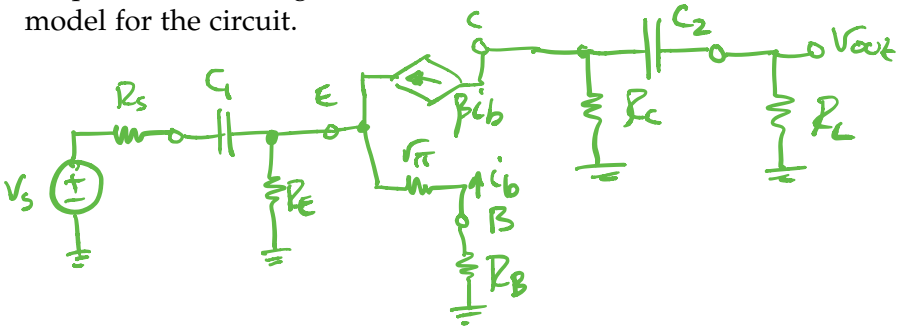
$V_C = 0.7V + (0.0207mA)(454\,k\Omega) = 10.1V$

$10.1V \leq V_C \leq 16.5V$

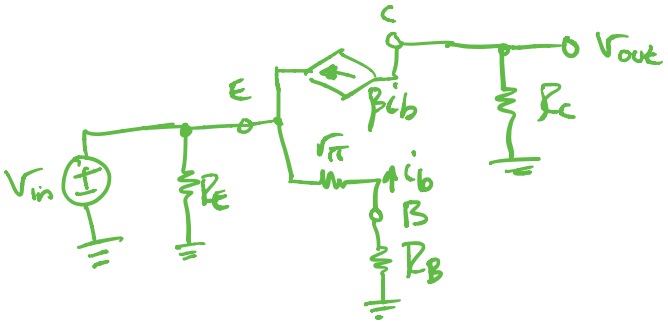
3. Use the following circuit to complete this problem.



(a) (10 points) Assuming that the transistor is in an active mode, sketch the small signal model for the circuit.



(b) (10 points) Write an expression for the *midband gain* of the circuit, $\frac{v_{out}}{v_{in}}$, with the load and source removed. The solution should be expressed in terms of resistors, β, r_π.



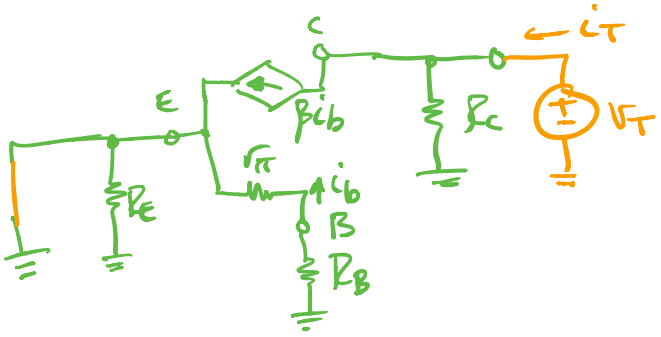
$$V_{out} = -\beta i_b R_C$$

$$V_{in} = -i_b(r_\pi + R_B)$$

$$\frac{V_{out}}{V_{in}} = \frac{\beta R_C}{r_\pi + R_B}$$

(c) (10 points) Find expression for the input and output resistance for the *midband* with the source and load removed.

$$R_{in} = R_E \| \frac{v_{in}}{(\beta+1) i_b} = \boxed{R_E \| \left(\frac{r_\pi + R_B}{\beta+1} \right)}$$



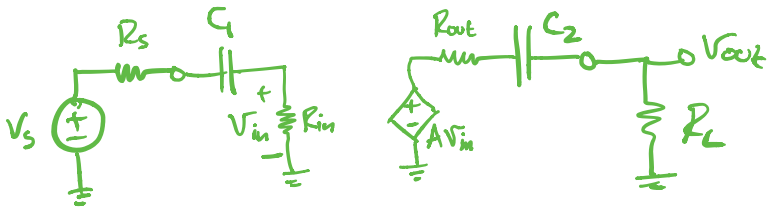
since $-i_b (r_\pi + R_B) = 0$

$i_b = 0$ and $\beta i_b = 0$

hence

$$\frac{v_T}{i_T} = \frac{v_T}{\frac{v_T}{R_C}} = \boxed{R_C = R_{out}}$$

(d) (10 points) Write expressions for the resistances seen by the capacitors C_1 and C_2.



$C_1 \rightarrow R_S + R_{in}$ $\quad$ $C_2 \rightarrow R_L + R_{out}$

(e) (5 points) Make a rough sketch of a possible Bode magnitude plot for this circuit.

